HISTORY AND CULTURE · MATHEMATICAL MODELLING

# The Languages of Europe

*When does a language become a national language?*

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How large must a group be, and how long must it live in a place, before its language becomes the language of the land? Abrams and Strogatz's differential equation for language competition gives a surprisingly clear answer: there is no fixed size, but a threshold at which size and status can substitute for each other. Here the model is extended to many languages on a map of Europe and applied to the centuries after the fall of Rome, to England after 1066, and to the nation states from the French Revolution to the present day.

## 01 Numbers are not enough

Europe has around 200 languages, but only a few dozen are national languages. What decides the matter is rarely how many people speak a language. It is what the language gives access to: the courts, the church, the school, the army, the market and the newspaper. The linguist Max Weinreich put it this way: a language is a dialect with an army and a navy.

History is full of examples. The Franks conquered Gaul and held the crown, yet their descendants speak French, a Romance language. The Normans conquered England, yet their descendants speak English. The Anglo-Saxons were also a minority in Britain, and still Brittonic Celtic died out across most of England. The question is what separates these three cases.

## 02 The Abrams–Strogatz model

In 2003 Daniel Abrams and Steven Strogatz proposed a simple model for two languages, X and Y, in the same population [1]. Let $x$ be the fraction speaking X. People switch from Y to X with a probability that grows with the number of X speakers and with the status $s$ of X:

$$\frac{dx}{dt} = c \cdot \left[ (1 - x) \cdot x^{a} \cdot s - x \cdot (1 - x)^{a} \cdot (1 - s) \right]$$

Here $c$ is a rate constant and $a$ an exponent, which they found to be $a \approx 1.31$ by fitting the model to historical data for Welsh, Scottish Gaelic, Quechua and other endangered languages. That $a$ exceeds 1 means a small language is disproportionately unattractive: it is rarely worth learning a language that almost nobody around you speaks.

## 03 The critical status

The model has only two stable states: everyone speaks X, or everyone speaks Y. Between them lies an unstable equilibrium. Setting $\mathrm{d}x/\mathrm{d}t = 0$ and dividing by $x(1 - x)$ gives $(1 - s)(1 - x)^{a-1} = s\,x^{a-1}$, i.e. the status $s^*$ that a group with fraction $x$ needs at least in order to grow:

$$\frac{s^*}{1 - s^*} = \left(\frac{1 - x}{x}\right)^{a - 1}$$

This answers the original question. A group making up half the population wins with slightly more than half of the status. A group of 1 % needs about four fifths of the status, and must

keep it for generations. The equilibrium is unstable: if the group lies slightly above the curve it grows, and its status requirement falls further. If it lies slightly below, the opposite happens.

| Group share x | Critical status s* | Relative to the majority's status |
|---|---|---|
| 1 % | 0.81 | 4.2 times |
| 2 % | 0.77 | 3.3 times |
| 5 % | 0.71 | 2.5 times |
| 10 % | 0.66 | 2.0 times |
| 20 % | 0.61 | 1.5 times |
| 30 % | 0.57 | 1.3 times |
| 50 % | 0.50 | 1.0 times |

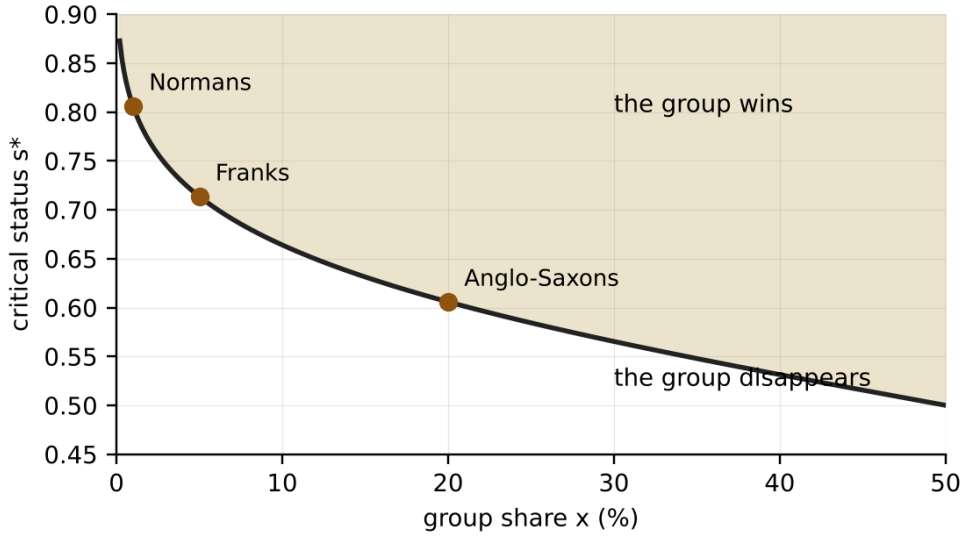



Figure 1. Critical status s* as a function of the group's share x for a = 1.31. Above the curve the group wins; below it, its language disappears. The points mark roughly estimated shares for three historical immigrant groups.

The time scale is $1/c$. With $c \approx 0.035$ per year this is about 30 years, roughly one generation. A language shift in a region typically takes three to five generations.

## 04 The model on a map of Europe

The interactive version at mhgfysik.dk extends the model in three ways. First, it applies to many languages at once:

$$\mathrm{d}x_j/\mathrm{d}t = c \cdot \left( x_j^{\,a}\, s_j - x_j \cdot \Sigma_i\, x_i^{\,a}\, s_i \right)$$

For two languages this is exactly the Abrams–Strogatz equation, and the fractions always sum to 1. Second, status is no longer a number one guesses. It is computed for every language, place and year as a weighted sum of historical factors:

$$s_j \propto 1 + \Sigma_f\, w_f \cdot F_{f,j}(\mathrm{place}, \mathrm{year})$$

$F$ records, for example, that Latin has the church in Gaul from 350, or that French has the school in Brittany from 1882. The 1 is the popular base every language has because somebody

speaks it at home. The weights $w$ can be set by the reader with sliders. Third, Europe is divided into a grid of cells some 30–45 km across, which exchange a little population with their neighbours. Immigration, conquest and expulsion enter as events at given places and times.

| Period | Status factors | Languages | Main events |
|---|---|---|---|
| 350–1000 | State & administration, church & writing, elite & army, trade & towns | Latin, Greek, Germanic, Celtic, Basque, Slavic, Arabic, Berber, other | Migration period, Anglo-Saxons, Franks, Slavs, the Arab conquest of 711, Magyars 895, Normans 911 |
| 1066–1500 | Crown & administration, law & literature, nobility & army, trade & towns | English, French, Celtic, Norse | The Conquest of 1066, Scottish burghs, Pembroke 1108, Ireland 1169, Wales 1282 |
| 1789–2026 | State & administration, school, media & church, work & economy | 16 languages, including standard and regional languages and newer immigrant languages | The Irish famine, border changes, the expulsions of 1945, guest workers, inflow to large cities |

The default values are $a = 1.31$, $c = 0.035$ per year and an exchange with neighbouring cells of 1 % per year. In the modern scenario a small exposure through school and media is added, so that a language can be learnt even when the neighbours do not speak it.

## 05 The Roman Empire and the migration period

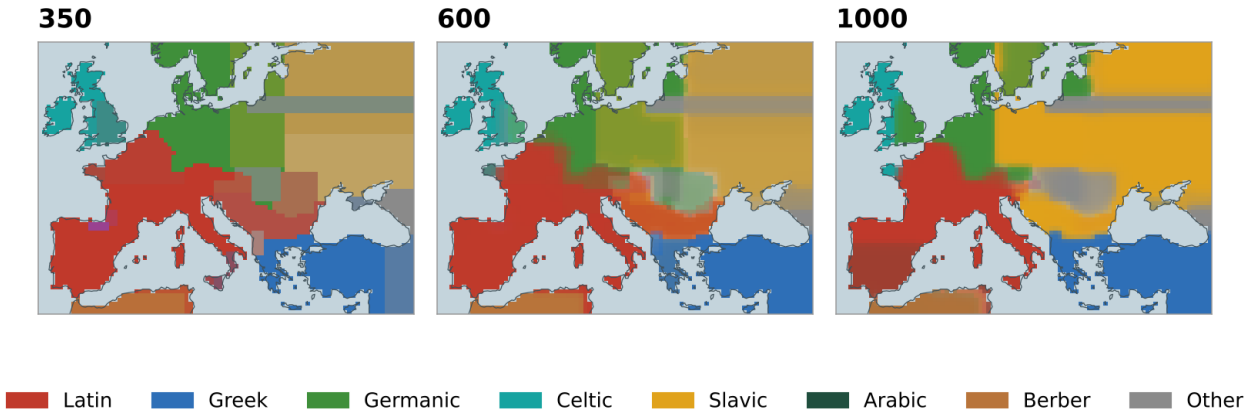



Figure 2. The model's language map in 350, 600 and 1000. Each cell is coloured by its largest language, blended with the second largest. Coastlines from Natural Earth.

Franks, Visigoths and Lombards took the crown in Gaul, Spain and Italy, but they were few, perhaps a few per cent. Latin had the church, writing, the towns and the majority. Germanic status never rose above the threshold, and within a few centuries the elite spoke Romance. Only along the Rhine, in Flanders and in Alsace, where settlement was dense, did the language border move. The Normans in Normandy after 911 are the same story in miniature: Norse vanished within a century.

In Britain the Roman institutions disappeared almost entirely after 410. Latin and Brittonic were left with only their popular base, while the Anglo-Saxons had both power and a larger share of the population. Here the Germanic language won. In the east the model shows Slavic spreading across the Balkans, while the Bulgars, a Turkic elite, become Slavic-speaking. The Magyars in Pannonia had both many settlers and the state from around the year 1000, and Hungarian survived.

## 06 1066 and all that

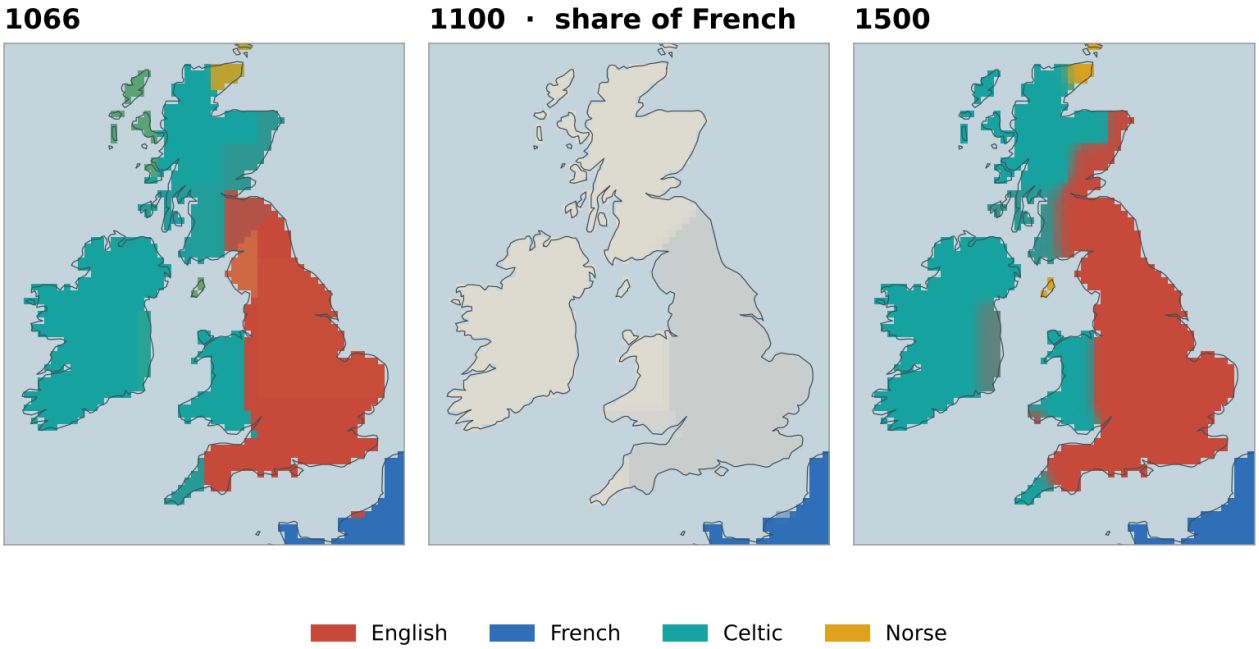



Figure 3. The British Isles in 1066 and 1500. The middle map shows only the share of French in 1100: French is visible as a thin layer over all of England, but never becomes a majority language.

The Normans were around 1 % of England's population, but held the crown, the nobility, the courts and, after 1100, literature. That gives a status just below the threshold of about 0.8. Then it fell: Normandy was lost in 1204, the Hundred Years' War made French the language of the enemy, and from 1362 court pleadings were in English (the Statute of Pleading). Henry IV, king from 1399, was the first monarch since the Conquest whose mother tongue was English.

What remained was English grammar with about 30 % of its words of French origin [8], for example *cow/beef* and *pig/pork*: the animal in the field kept its English name, while the meat on the lord's table got a French one. The Danelaw is the same story on a smaller scale. Norse vanished as a spoken language, but left behind *they, them, sky, egg* and *take*. With the model's slider for group size one can explore how many more Normans it would have taken to make England French-speaking.

14 October 2026 marked 960 years since the Battle of Hastings, and until 11 July 2027 the Bayeux Tapestry is on show at the British Museum in London. The tapestry shows Halley's Comet, which appeared in April 1066, and the Norman ships: clinker-built longships with square sails and dragon heads, of the same type as the Skuldelev ships from Roskilde Fjord in Denmark. The Normans had kept the Viking art of shipbuilding, but lost their Norse language within a couple of generations after 911. The technology survived; the language did not.




Picture 1. Halley's Comet over England in April 1066. The inscription ISTI MIRANT STELLAM means "these men marvel at the star". On the right the newly crowned King Harold receives the news, and in the lower border empty ships foreshadow the invasion. Bayeux Tapestry, 1070s. Photo: Myrabella, Wikimedia Commons (public domain).




Picture 2. The fleet crosses the Channel (scenes 38–39). The inscriptions read ET VENIT AD PEVENESAE ("and came to Pevensey"), HIC EXEUNT CABALLI DE NAVIBUS ("here the horses leave the ships") and ET HIC MILITES FESTINAVERUNT HESTINGA ("and here the soldiers hurried to Hastings"). Bayeux Tapestry, 1070s. Photo: Myrabella, Wikimedia Commons (public domain).

The inscriptions are neither in French nor in English, but in Latin. For a long time the written language was a third language that nobody spoke at home, while the vernaculars competed for speakers. That competition is exactly what the model describes.

## 07 The nation state and immigration

In 1789 only a minority in France spoke French as their everyday language. Through the state, conscription, school (the Ferry laws of 1882), newspapers, radio and television the standard language became almost universal [6], and Occitan and Breton collapsed during the twentieth century. The same happened to Low German in northern Germany. Italian is the opposite case: at unification in 1861 only a few per cent spoke standard Italian [7], and it took school, radio and above all television after 1954 to make it the language of the majority. Catalan, Welsh and Basque fare better than Occitan and Breton because they gained institutions of their own: schools, television and self-government.

In the model, newer immigrant languages in the large cities have low institutional status. As long as new speakers arrive, their share settles at a level where inflow and language shift balance. With the default values this is a few per cent in the large cities and below one per cent of the whole map. Studies typically find a shift to the majority language over three generations [9, 10], but with clear exceptions: a recent French study found robust maintenance of Turkish even in the third generation [9]. The model therefore does not say that a language cannot survive. It says that a language without institutions of its own persists only as long as it is renewed from outside or carried by close-knit networks, and that becoming a national language takes far more than numbers.

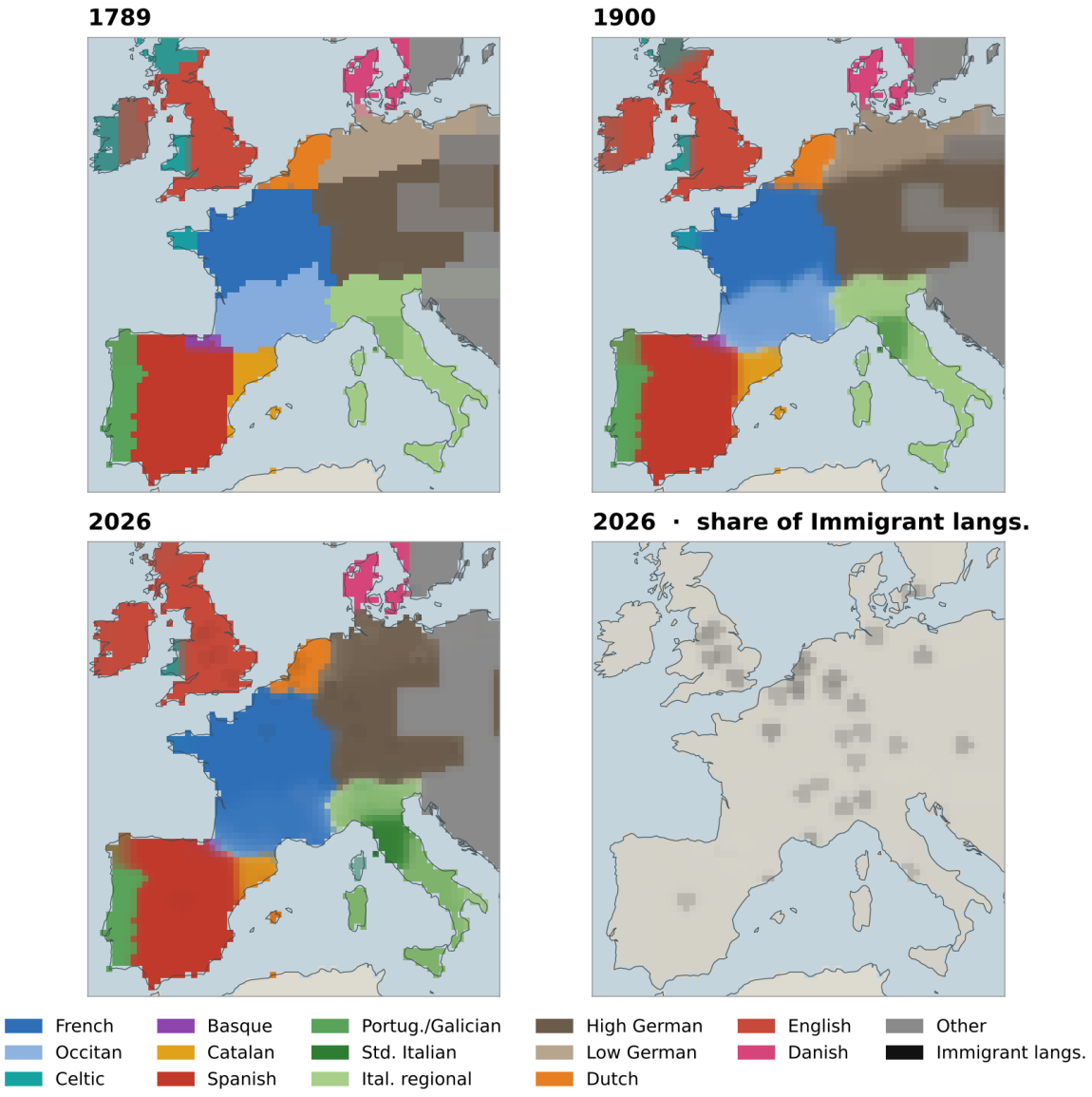



Figure 4. Western Europe in 1789, 1900 and 2026, and the share of newer immigrant languages in 2026 with the default inflow (0.3 % of a large city's population per year from 1955). The share is shown on a square-root scale so that small shares are visible.

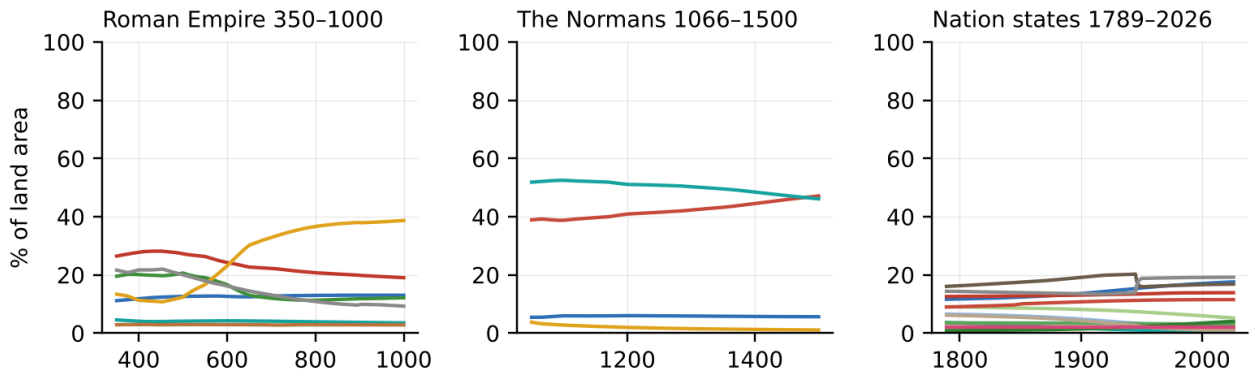



Figure 5. Each language's share of the map's land area over time in the three scenarios. Note that shares are by area, not by population.

## 08 Model and reality

| Place | Model | History |
|---|---|---|
| Paris and Rouen, year 1000 | Romance almost 100 % | Romance; Frankish and Norse gone |
| London, year 1000 | Germanic almost 100 % | Old English |
| Western Brittany, year 1000 | Celtic dominant | Breton |
| Budapest, year 1000 | "Other" (Hungarian) dominant | Hungarian |
| England, year 1500 | French gone | English with French loanwords |
| Toulouse, 2026 | French c. 80 % | Occitan is a minority language of older people |
| Barcelona, 2026 | Catalan c. 2/3 | Catalan and Spanish side by side |
| Ceredigion, 2026 | Welsh c. 2/3 | About half speak Welsh |

The model reproduces the broad picture, but it has clear limitations:

- It treats people as monolingual. In reality a shift almost always passes through a bilingual generation, and a language can survive for a long time as a second language. Extensions with bilingualism exist [3, 4, 5].
- Shares are by area, not by population. A large city weighs no more than a cell of mountains.
- Initial distributions, borders and factors are rough estimates from standard works. The map shows mechanisms, not precise numbers.
- Romanian survived north of the Danube, which the model does not capture. It remains one of the puzzles of linguistic history.
- Immigration after 1955 is an illustrative inflow into 32 large cities. It is not demographic data.

## 09 Exercises

- Show by setting $\mathrm{d}x/\mathrm{d}t = 0$ that $s^*/(1 - s^*) = ((1 - x)/x)^{a-1}$. Why is the equilibrium unstable?
- What happens to the threshold if $a = 1$? What does that mean linguistically?
- How many times more Normans would have been needed for England to become French-speaking? Use the slider for group size.
- Set the weight on "Church & writing" to 0 in the Roman scenario. What happens to Latin in Gaul?

- How large an inflow is needed before an immigrant language becomes the majority language of a large city in the model? Which assumptions lie behind the answer?

## 10 References


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